

Hello Robot Stretch

Comprehensive GitHub Repository & Open-Source Code Research Report

With GazeBot Project Recommendations

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1. Executive Summary

Hello Robot Inc. maintains an extensive open-source software ecosystem for the Stretch family of mobile manipulators, hosted on GitHub under the organization github.com/hello-robot. With **34+ repositories**, the ecosystem spans low-level hardware drivers, ROS 1/2 integration, AI-powered manipulation, simulation, teleoperation, tutorials, and community contributions. Nearly all code is released under permissive licenses (Apache 2.0, MIT, LGPL).

This report provides a comprehensive analysis of every major repository, including architecture, key modules, APIs, dependencies, and capabilities. It also catalogs 30+ peer-reviewed publications from ICRA, IROS, CoRL, HRI, NeurIPS, CVPR, and RSS that leverage the Stretch platform. A dedicated section maps these repositories to the GazeBot project's needs -- a head-gaze + sip-and-puff shared autonomy system being developed at the UIUC LIFE Home.

2. GitHub Organization Overview

The hello-robot GitHub organization (github.com/hello-robot) hosts 34+ repositories. The key repositories are:

Repository	Description	Stars	License	Language
stretch_ai	AI behaviors, LLM agents, grasping, navigation	219	Apache 2.0	Python
stretch_ros2	ROS 2 Humble packages (Nav2, MoveIt2, SLAM)	111	Apache 2.0	Python
stretch_ros	ROS 1 Noetic packages (legacy)	185	Apache 2.0	Python
stretch_body	Python SDK for hardware control	--	LGPLv3	Python
stretch_firmware	Arduino firmware for custom PCBAs	--	GPLv3	C++
stretch_tutorials	40+ tutorials (Python, ROS 2)	15	--	Jupyter
stretch_dex_teleop	6-DOF dexterous teleoperation	--	--	Python
stretch_tool_share	Community grippers, tools, CAD files	34	Apache 2.0	Python
stretch_community	Community portal, papers, third-party repos	13	--	Markdown
stretch_install	Installation and setup scripts	--	--	Shell
stretch_factory	Factory calibration and debug tools	--	--	Python
stretch_diagnostics	Diagnostic and health check tools	--	--	Python
stretch_hardware_guides	Hardware documentation for all models	--	CC BY-NC-SA	Markdown
stretch_urdf	URDF models and meshes	--	--	URDF/Xacro
stretch_pyfunmap	Python FUNMAP nav+manipulation planner	--	--	Python
stretch_web_teleop	Web-based remote teleoperation (WebRTC)	--	--	JS/Python
stretch_mujoco	MuJoCo simulation stack	--	BSD 3-Clause	Python
stretch_isaacsim	NVIDIA Isaac Sim integration	--	Apache 2.0	Python
stretch_forcesight	Text-guided manipulation (ICRA 2024)	--	--	Python
stretch_visual_servoing	Visual servoing via gripper camera	--	--	Python
stretch_docking	Autonomous docking via visual servoing	--	--	Python

3. stretch_ai -- AI & Intelligent Behaviors

Repository: github.com/hello-robot/stretch_ai | **Stars:** 219 | **Forks:** 47 | **License:** Apache 2.0 (+ MIT for Meta HomeRobot components)

Lead: Chris Paxton, Senior Embodied AI Lead at Hello Robot (prev. NVIDIA Research, Meta FAIR)

Stretch AI is the flagship software framework for building intelligent robot behaviors on Stretch 3. It provides integrated tools for grasping, manipulation, 3D mapping, navigation, LLM-based task planning, speech processing, and embodied question answering.

3.1 Architecture

Distributed client-server design: The robot runs a ROS 2 environment with a ZMQ-enabled server. A separate GPU workstation (connected via WiFi) runs compute-intensive AI models. Communication happens over a ZMQ network bridge.

Directory	Purpose
src/stretch/agent/	Robot agent orchestration, ZMQ client, manipulation operations, task FSMs
src/stretch/app/	22+ runnable applications (ai_pickup, grasp_object, mapping, chat, etc.)
src/stretch/perception/	Vision & object detection (OWLv2, SAMv2, YOLO)
src/stretch/mapping/	3D voxel mapping & DynaMem spatio-semantic memory
src/stretch/motion/	Motion planning (A*, RRT, RRT-Connect)
src/stretch/llms/	LLM clients (Qwen2.5, OpenAI GPT-4o, Gemma, Gemini)
src/stretch/audio/	Whisper speech-to-text, text-to-speech
src/stretch/simulation/	MuJoCo simulation integration
src/stretch_ros2_bridge/	ROS 2 to ZMQ bridge for AI workstation communication

3.2 Core API -- RobotClient

```
from stretch.agent import RobotClient
robot = RobotClient(robot_ip='192.168.1.15')
robot.move_to_nav_posture()
robot.move_base_to([0.5, 0, 0], relative=True)
robot.move_to_manip_posture()
robot.arm_to([0.1, 0.5, 0, 0, 0, 0])
```

3.3 AI Capabilities

Capability	Details
Navigation & Mapping	Frontier-based/value-based exploration, 3D voxel maps from RGB-D, A*/RRT planning, open-vocab object localization
Manipulation	Language-directed pick-and-place, visual servoing, AnyGrasp 6-DOF grasps, OWLv2+SAMv2 target selection
Voice & Language	OpenAI Whisper STT, TTS feedback, LLM command parsing (Qwen2.5/GPT-4o), multi-turn chat
DynaMem	Dynamic spatio-semantic memory, handles moving objects, 70% pick-and-drop success on non-stationary objects

Capability	Details
Learning from Demo	ACT, VQ-BeT, Diffusion Policy architectures via HuggingFace LeRobot, ~50 episodes recommended
Embodied QA	Robot explores to answer questions, uses VLMs + Gemini for multimodal reasoning
LLM Agent	Finite state machines with operations: pickup(), explore(), place(), say(), wave(), find()

3.4 Key Applications

Application	Command	Description
AI Pickup	<code>python -m stretch.app.ai_pickup --use_llm --use_voice</code>	Language-directed pick-and-place with voice
Mapping	<code>python -m stretch.app.mapping --explore-iter 10</code>	Autonomous 3D environment mapping
Grasp Object	<code>python -m stretch.app.grasp_object --target '...'</code>	Single object grasping by name
DynaMem	<code>python -m stretch.app.run_dynamem</code>	Dynamic semantic memory system
Voice Chat	<code>python -m stretch.app.chat --voice</code>	Conversational LLM agent
Keyboard Teleop	<code>python -m stretch.app.keyboard_teleop</code>	WASD robot control
Dex Teleop	<code>python -m stretch.app.dex_teleop.ros2_leader</code>	Demonstration data collection
EQA	<code>python -m stretch.app.run_eqa</code>	Embodied question answering

3.5 Dependencies

Hardware: Stretch 3, Ubuntu 22.04 workstation with NVIDIA GPU (tested RTX 4090), dedicated WiFi AP.

Key Python: torch>=2.6, ultralytics (YOLO), openai-clip, transformers>=4.50, openai>=1.88, pin (Pinocchio IK), rerun-sdk, librosa, PyAudio, openai-whisper, open3d, hydra-core.

Docker: Available as hellorobotinc/stretch-ai_cuda-11.8:latest

4. stretch_ros2 -- ROS 2 Packages

Repository: github.com/hello-robot/stretch_ros2 | **Stars:** 111 | **Forks:** 65 | **Branch:** humble | **ROS 2:** Humble Hawksbill (Ubuntu 22.04)

4.1 All Packages (11 total)

Package	License	Description
stretch_core	Apache 2.0	Core ROS 2 drivers, hardware interface via stretch_body SDK. Modes: position, navigation, trajectory, homing, stowing, runstopped
stretch_description	BSD 3-Clause	URDF/xacro robot description files and STL meshes
stretch_nav2	Apache 2.0	Nav2 navigation stack (SLAM via slam_toolbox, localization via AMCL, path planning)
stretch_calibration	GPLv3	Generates calibrated URDF models unique to each physical robot
stretch_deep_perception	Apache 2.0	Deep learning perception demos (face, object, body detection)
stretch_demos	Apache 2.0	Autonomous manipulation demonstrations (hello_world, open_drawer, clean_surface, handover)
stretch_funmap	LGPLv3	Fast Unified Navigation, Manipulation And Planning tools
stretch_octomap	Apache 2.0	3D probabilistic mapping using octrees
stretch_rtabmap	Apache 2.0	Real-Time Appearance-Based 3D Mapping and visual navigation
stretch_simulation	Apache 2.0	MuJoCo-based simulation (not Gazebo), supports RoboCasa kitchen environments
hello_helpers	Apache 2.0	Shared Python utilities (HelloNode class, move_to_pose, plane fitting)

4.2 Key Nodes (stretch_core)

Node	Description
stretch_driver.py	Core node communicating with stretch_body. Modes: position, navigation, trajectory, homing, stowing, runstopped
joint_trajectory_server.py	Action server for MoveIt2 and other trajectory planners
keyboard_teleop.py	Keyboard control with integrated demo triggers
detect_aruco_markers.py	ArUco marker detection and pose estimation
d435i_*.py	Intel RealSense D435i 3D camera nodes (high/low resolution)

4.3 Key Launch Files

Launch File	Package	Purpose
stretch_driver.launch.py	stretch_core	Main driver + robot_state_publisher + joint_state_publisher
offline_mapping.launch.py	stretch_nav2	SLAM-based mapping with slam_toolbox
navigation.launch.py	stretch_nav2	Autonomous navigation with Nav2 + AMCL
visual_mapping.launch.py	stretch_rtabmap	3D RGBD-based environment mapping
visual_navigation.launch.py	stretch_rtabmap	Visual navigation in pre-mapped spaces

Launch File	Package	Purpose
stretch_mujoco_driver.launch.py	stretch_simulation	MuJoCo simulation driver
display.launch.py	stretch_description	Quick URDF visualization in RViz

4.4 Nav2 Integration

Uses **slam_toolbox** for SLAM mapping, **AMCL** for localization in pre-mapped environments, RPLidar LaserScan for obstacle avoidance. Maps saved as .pgm + .yaml at \${HELLO_FLEET_PATH}/maps/. Recovery behaviors include 180-degree spin or backup. Supports programmatic goals via **Nav2 Simple Commander API**.

4.5 MoveIt2 Integration

Three planning groups: **stretch_arm**, **stretch_gripper**, **stretch_head**. Mobile base adds 2 additional DOF for whole-body planning (arm + base simultaneously). Uses joint_trajectory_server.py action server for execution.

4.6 Key Topics & Services

Type	Name	Description
Topic	/mode	Current driver mode (15 Hz)
Topic	/battery	Battery state (voltage, current)
Topic	/is_homed	Encoder homing status
Service	/home_the_robot	30-second homing sequence
Service	/stow_the_robot	Stow arm into base footprint
Service	/stop_the_robot	Immediate stop
Service	/runstop	Programmatic emergency stop (SetBool)
Service	/funmap/trigger_*	FUNMAP actions (head_scan, drive_to_scan, reach_until_contact, etc.)

5. stretch_ros -- ROS 1 Legacy Packages

Repository: github.com/hello-robot/stretch_ros | **Stars:** 185 | **Forks:** 95 | **ROS:** Noetic (EOL May 2025) | **12 packages**

Aspect	stretch_ros (ROS 1)	stretch_ros2 (ROS 2)
Navigation	move_base + gmapping	Nav2 + slam_toolbox
Manipulation	Not integrated	Movel2 (whole-body planning)
Simulation	Gazebo Classic (ros_control)	MuJoCo (+ RoboCasa kitchens)
Driver Modes	position, navigation	position, navigation, trajectory (new)
Dashboard	GUI dashboard included	No dashboard
Build System	catkin	colcon
Communication	TCPROS	DDS (with QoS profiles)
Status	Mature but EOL	Active development, LTS until May 2027

Note: For new projects (including GazeBot), stretch_ros2 is strongly recommended. stretch_ros is documented here for completeness and backward compatibility reference.

6. stretch_body -- Python Hardware SDK

Repository: github.com/hello-robot/stretch_body | **License:** LGPLv3

Packages: hello-robot-stretch-body (core) + hello-robot-stretch-body-tools (CLI utilities)

6.1 Class Hierarchy

```
Device (base class)
+-- Robot (top-level orchestrator)
+-- Stepper (stepper motor interface)
+-- PrismaticJoint -> Arm, Lift
+-- Base (differential drive)
+-- Pimu (power/IMU board)
+-- Wacc (wrist accelerometer)
+-- DynamixelChain -> Head, EndOfArm
+-- DynamixelHelloXL430 -> WristYaw, WristPitch, WristRoll, StretchGripper
```

6.2 Key API Patterns

Joint Type	Command Methods	Execution Model
Arm, Lift, Base (steppers)	move_to(), move_by(), set_velocity()	Queued: push_command() then wait_command()
Head, Wrist, Gripper (Dynamixels)	move_to('joint', val), set_velocity(v)	Immediate execution

```
import stretch_body.robot
robot = stretch_body.robot.Robot()
robot.startup()
robot.arm.move_to(0.3) # meters
robot.lift.move_to(0.5) # meters
robot.push_command()
robot.wait_command()
robot.head.move_to('head_pan', 0.5) # radians, immediate
robot.end_of_arm.move_to('stretch_gripper', 50) # percent open
```

6.3 Sensor Access

Sensor	Access Path	Data
Joint Encoders	robot.arm/lift/base.status['pos']	Position (m/rad), velocity, force/effort
Base IMU (BNO085)	robot.pimu.status	9-DOF: accel, gyro, magnetometer
Base Odometry	robot.base.status['x'/y'/theta']	X, Y (meters), heading (radians)
Cliff Sensors (4x IR)	robot.pimu.status	Floor drop-off detection
Battery	robot.pimu.status	Voltage, current, charger state
Wrist Accel (ADXL343)	robot.wacc.status	3-axis accel, tap/bump detection
Wrist Analog/Digital I/O	robot.wacc.status	a0 (analog), d0-d3 (digital)

6.4 Motion Control

Position control: Absolute (move_to) and relative (move_by) for all joints. **Velocity control:** set_velocity() for continuous motion. **Trajectory following:** Cubic-spline interpolation with waypoints (time, position, velocity). **Guarded contact:** Current-sensing halts motion on effort threshold (protects against collisions). **Collision avoidance:** RobotCollision manager dynamically limits joint ranges at ~10 Hz. **Units:** SI throughout -- meters,

radians, seconds, Newtons.

6.5 Command-Line Tools

Tool	Purpose
stretch_robot_home.py	Calibrate/home all joints after boot
stretch_robot_stow.py	Return arm/tool to safe position
stretch_robot_system_check.py	Scan all hardware, validate bus connections
stretch_robot_keyboard_teleop.py	Keyboard teleoperation
stretch_xbox_controller_teleop.py	Xbox gamepad teleoperation
stretch_robot_jog.py	Interactive jogging of all joints
stretch_robot_battery_check.py	Battery voltage/current status
REx_firmware_updater.py	Firmware update utility

7. stretch_firmware -- Arduino Firmware

Repository: github.com/hello-robot/stretch_firmware | **License:** GPLv3

MCU: Atmel SAMD21G18A-AUT (ARM Cortex-M0+, 32-bit, 48 MHz)

7.1 Firmware Components

Component	Files	Purpose
hello_stepper	30 files	Stepper motor controller: closed-loop current control, trapezoidal trajectory generation, multi-motor sync, guarded contact
hello_pimu	27 files	Power/IMU board: BNO085 IMU, cliff sensors, battery monitoring, runstop logic, sync pulse generation, buzzer/LED control
hello_wacc	11 files	Wrist accelerometer: ADXL343 driver, tap/bump detection, analog/digital I/O

7.2 Communication Protocol

Uses **COBS** (Consistent Overhead Byte Stuffing) for zero-free packet framing over USB serial, with **CRC16** for data integrity. Custom RPC protocol with packed C structs. Protocol versions P0-P6 supported. Asyncio support for non-blocking stepper communication.

7.3 PIMU Safety Monitoring (100 Hz)

Monitor	Sensor	Threshold/Action
Voltage	Battery ADC	Low alert at 10.5V; runstop on critical low
Current	Current sensor / e-fuse	Alert at 6A; overcurrent protection
Tilt	IMU accelerometer	Tilt detection triggers runstop
Cliff	4x Sharp IR sensors	Floor drop-off triggers runstop
Watchdog	Timer	11ms timeout, automatic reset on hang

8. stretch_tutorials -- Educational Resources

Repository: github.com/hello-robot/stretch_tutorials | **Stars:** 15 | 469 commits

Organized into three progressive tracks with 40+ tutorials.

8.1 Tutorial Tracks

Track	Focus	Key Topics
Track A: Developing with Stretch	Foundational	CLI tools, networking, system startup, basics
Track B: Python Tutorials	stretch_body API	Motion control, sensors (RealSense), audio, inverse kinematics, visual servoing
Track C: ROS 2 Tutorials (40+)	ROS 2 integration	11 basics + 6 advanced + 12 examples (see below)

8.2 ROS 2 Tutorial Topics

Basics (11): Packages/Nodes, Simulation, Teleop, RViz, Joint Trajectory, HelloNode, Robot Driver, Twist Control, Sensors, Nav2 Stack, Deep Perception.

Advanced (6): ArUco Detection, Offloading Computation, Race Conditions, Autonomy Demos, FUNMAP, Perception.

Examples (12): Voice-to-Text, Voice Teleop, Laser Scan Filtering, ArUco Localization, Print Joint States, Store Effort, Tf2 Broadcaster/Listener, Image Capture, Collision Avoidance, Obstacle Avoider.

9. stretch_dex_teleop -- Dexterous Teleoperation

Enables 6-DOF dexterous teleoperation of Stretch 3 using modified kitchen tongs with ArUco visual markers tracked by a webcam. Translates hand motions into target poses + gripper commands at 15+ Hz.

9.1 Capabilities

- 6-DOF end-effector control: position via IK, orientation via wrist joints, grip via gripper
- Single-arm and bimanual operation (two Stretch 3 robots simultaneously)
- Slow mode (default, conservative) and Fast mode (maximum speed, reduced contact sensitivity)
- Data collection platform for robot learning -- integrated with HuggingFace LeRobot
- Multiprocessing: sensor acquisition decoupled from motor control via shared memory

9.2 Hardware Kit (\$295)

Logitech C930e webcam (1080p), LED ring light, adjustable stand/tripod, ArUco tongs, Stretch 3 with dex wrist. Bimanual setup requires two kits with distinct ArUco marker IDs.

10. stretch_tool_share -- Community Tools & Accessories

Repository: github.com/hello-robot/stretch_tool_share | **Stars:** 34 | **21 tools**

Category	Tools
Computing & Sensing	NVIDIA Jetson Orin AGX mount, wrist USB board camera
Dexterous Manipulation	Dex Wrist (RE1/RE2 add-on), Dex Wrist Beta, ReactorX wrist
Accessories	Phone holder, dry erase holder, Swiffer mount, tray/cup holder, button pusher, puller, card holder, arm tray, swivel tablet mount, adapted spoon
Infrastructure	Docking station, teleop kit (2x fish-eye cameras), shoulder joint
Reference Models	Stretch 2 STEP, RE1 arm/head models

Design files include STL (3D printing), STEP (CAD), URDF+meshes (ROS), and Gazebo support. Robot components: CC BY-NC-SA 4.0. Tool accessories: Apache 2.0.

11. stretch_community -- Community Portal

Curated portal of third-party projects, publications, simulations, and educational resources.

11.1 Featured Community Projects

Organization	Project	Description
Meta AI + CMU	Semantic Navigation	Find object instances in unseen home environments
UW HCR Lab	stretch-web-app	Enhanced web teleoperation interface (Prof. Maya Cakmak)
Meta AI	Fairo	Unified robotics platform (led by Soumith Chintala)
Avanade	emtech-stretch-labs	NextMind BCI, Azure IoT, DroneDeploy integration
Johns Hopkins	icl_stretch_ros	Tool retrieval for hospital sterile processing
UMass Lowell	stretch_gui / stretch_uml	Semi-autonomous teleoperation interfaces

11.2 Simulation Environments

Source	Platform	Details
U. Washington	Gazebo	Improved indoor environment + object models
Meta AI (Chris Paxton)	Habitat2	Training home assistant behaviors
CMU (Prof. Erickson)	OpenAI Gym	assistive-gym -- assistive robotics tasks
Cardiff University	Gazebo Harmonic	ROS 2 integration

12. Additional Repositories

12.1 stretch_install

License: GPLv3 | **Language:** Shell (96.1%), Python (3.9%)

Automates software setup for Stretch robots across three installation paths:

Script	Purpose	Runtime
stretch_new_robot_install.sh	Full OS + software stack deployment on fresh robot	20-30 min (wired)
stretch_new_user_install.sh	Provisions new Ubuntu user account with all Stretch packages	~10 min
stretch_update_ros_workspace.sh	Creates/recompiles ROS1 catkin or ROS2 ament workspaces	~5 min

Supported Ubuntu versions: 18.04 (Melodic), 20.04 (Noetic), 22.04 (Humble).

Requirements: Wired USB keyboard/mouse (Bluetooth incompatible with BIOS), monitor, 2 USB flash drives (one >8GB), Ethernet recommended. Robot model codes: 're1' (RE1), 're2' (Stretch 2), 'se3' (Stretch 3).

12.2 stretch_factory

License: Proprietary (U.S. Patent 11,230,000) | **Install:** pip install hello-robot-stretch-factory

Low-level factory calibration and debug tools (all prefixed with **REx_**), intended for use under Hello Robot support guidance.

Category	Tools
Base & IMU Calibration	REx_base_calibrate_imu_collect/process, REx_base_calibrate_wheel_separation
Dynamixel Management	REx_dynamixel_id_scan/change/jog/reboot/set_baud, REx_hello_dynamixel_jog
Stepper Operations	REx_stepper_jog/gains/ctrl_tuning/calibration_run/type, flash-to-YAML and YAML-to-flash calibration export
Sensor Calibration	REx_calibrate_guarded_contact, REx_calibrate_range, REx_cliff_sensor_calibrate, REx_gripper_calibrate, REx_wacc_calibrate
System & Firmware	REx_firmware_updater, REx_discover_hello_devices, REx_usb_reset, REx_D435i_check

Includes 18 documented factory update procedures (001-018) covering hardware modifications, servo swaps, and calibration sequences.

12.3 stretch_diagnostics (ARCHIVED Feb 2026)

Status: Archived -- superseded by stretch_system_check.py in stretch_body.

Install: pip3 install hello-robot-stretch-diagnostics

CLI tool **stretch_diagnostic_check.py** with test suite flags:

Flag	Tests	Count
--simple	Accessories, filesystem, firmware, params, PIMU, RealSense, RPLidar, software, steppers, udev, USB, wacc, wifi	17
--power	Battery health, battery loading, charger	3
--realsense	Cable integrity, frame rate	2
--stepper	Calibration data match, power, runstop, sync	4
--dynamixel	Zero position, hardware, efforts, range of motion	4

Flag	Tests	Count
--ros	Calibration, sourced distro, URDF	3
--cpu	CPU usage and temperature	1
--arm / --lift	Effort through range of motion	1 each

12.4 stretch_hardware_guides

Hardware documentation for all Stretch models. Key documents: Safety Guide, Battery Maintenance Guide, Hardware Guide, and Dex Wrist Guide (per model). Published under CC BY-NC-SA.

Feature	RE1	RE2 / Stretch 2	Stretch 3
Arm Material	Carbon fiber	Carbon fiber	Aluminum
Base IMU	FXOS8700+FXAS21002	Same	BNO085 (9 DOF)
Head Camera	D435i	D435i	D435if + Arducam B0385 (140° FOV)
Gripper Camera	None	None	Intel RealSense D405
Dex Wrist	Optional add-on	Optional add-on	Built-in (3 DOF standard)
Head Tilt Servo	XL430-W250-T	XL430-W250-T	XC430-W240-T (upgraded)
Lift Brake	None	None	Yes (engages on power loss)
Battery	2x 12V AGM SLA, 18AH	Same	Same (2-5 hr runtime)

12.5 Other Notable Repositories

Repository	Description
stretch_urdf	URDF models and meshes for all Stretch variants, source assets for stretch_description
stretch_pyfunmap	Python FUNMAP: combined navigation + manipulation using geometric models and computer vision
stretch_web_teleop	Remote web teleoperation via browser (ROS2, WebRTC, Nav2, TypeScript). Works locally or over internet.
stretch_mujoco	MuJoCo simulation: position control (arm/head/gripper), velocity control (base), calibrated camera RGB+depth, 2D lidar. ROS2 compatible with Nav2/Web Teleop. BSD 3-Clause.
stretch_isaacsim	NVIDIA Isaac Sim integration for Stretch (Apache 2.0)
stretch_forcesight	Pre-trained ForceSight model for text-guided manipulation with visual-force goals (ICRA 2024)
stretch_visual_servoing	Visual servoing using Stretch 3's gripper D405 camera
stretch_docking	Autonomous docking demo via visual servoing
stretch_deep_perception_models	Pre-trained deep learning models for perception (faces, objects, bodies)
stretch_show_tablet	Tablet interface for Stretch
stretch_ctrl_dev	Experimental controls research
lerobot (fork)	Hello Robot's fork of HuggingFace LeRobot for imitation learning (stretch-act branch)
ros2_numpy	Tools for converting ROS messages to/from numpy arrays (MIT)

13. Academic Publications & Research Papers

The Stretch platform has generated 30+ peer-reviewed publications at top venues. Below are the most significant.

13.1 Foundational Papers (Hello Robot Team)

Title	Authors/Affiliation	Venue	Key Contribution
The Design of Stretch	Kemp, Edsinger, Clever (Georgia Tech / Hello Robot)	ICRA 2022	Core design paper: 23 kg, <\$20K, 8 prototype iterations, home deployment evaluation
ForceSight	Collins, Houff, Tan, Kemp	ICRA 2024	Text-guided manipulation with visual-force goals, 81% success on 10 household tasks
Robots for Humanity	Ranganeni, Nguyen, Evans (UW/Hello Robot)	HRI 2024	In-home deployment with Henry Evans (quadriplegia), 3 one-week participatory sessions
Immersive Participatory Design	Olatunji, Nguyen, Cakmak, Kemp, Rogers, Mahajan	Ergonomics 2024	4-week in-home deployment, 68% performance increase, 72% satisfaction improvement

13.2 Major Conference Papers

Title	Venue	Key Result
Harmonic Mobile Manipulation	IROS 2024 (Best Paper)	End-to-end nav+manipulation, 17.6% improvement, sim-to-real transfer on Stretch
PoliFormer	CoRL 2024 (Outstanding Paper)	RGB-only navigation, 85.5% success, 28.5% improvement, zero-shot sim-to-real
HomeRobot OVMM	NeurIPS 2023 Challenge	Open-vocab mobile manipulation benchmark, Stretch RE2 real-world stack
SPOC	CVPR 2024	Text instruction following, trained on 200K houses, zero-shot sim-to-real
OK-Robot	arXiv 2024	Zero-shot pick-and-drop, 58.5% success (1.8x prior SOTA), 82% uncluttered
Robot Utility Models (RUMs)	arXiv 2024 (NYU/Hello Robot/Meta)	Zero-shot deployment, 90% success in novel environments
VRB: Affordances from Videos	CVPR 2023 (CMU)	12 tasks learned from internet videos, 25-min adaptation per task
WHIRL	RSS 2022 (CMU)	Human-to-robot imitation, one-shot generalization across 20 tasks
Dobb-E: Bringing Robots Home	arXiv 2023 (NYU/Meta)	81% success with 5 min demos, 22 NYC homes, open-source
Open X-Embodiment (RT-X)	ICRA 2024 (34 labs)	Multi-robot dataset, 50% improvement across 5 robots, Stretch included

13.3 Assistive Robotics & Healthcare

Title	Venue	Key Contribution
Bodies Uncovered (blanket manipulation)	RA-L 2022 (CMU/GT)	Sim-to-real blanket manipulation on Stretch, uncovering body parts
RoBE (graph-based blanket dynamics)	RA-L 2023 (CMU)	Graph-based cloth dynamics, 12-person human study

Title	Venue	Key Contribution
HAT: Head-Worn Assistive Teleop	arXiv 2023 (CMU)	Head-worn interface for Henry Evans, week-long in-home study
Bimanual HD-EMG Control	arXiv 2026 (CMU)	12-day in-home study, quadriplegia user, muscle-signal control of Stretch
Accessible Remote Tele-operation	RO-MAN 2021 (UW)	Browser-based accessible interface, 18-participant study
AccessTeleopKit	UIST 2024 (UW)	Open-source toolkit for accessible web-based tele-operation
Robotic Support for Older Adults	Frontiers 2025	Feasibility assessment, participatory design, social connection

13.4 Key Research Institutions

Institution	Key Researchers	Focus
Allen Institute for AI (AI2)	Ehsani, Kembhavi, Weihs	Embodied AI, sim-to-real (SPOC, PoliFormer, HarmonicMM)
New York University	Pinto, Shafiullah	Imitation learning, home robots (Dobb-E, RUMs, VINN, OK-Robot)
Carnegie Mellon University	Pathak, Bahl, Erickson	Human video imitation, assistive robotics (WHIRL, VRB, HAT, EMG)
Georgia Tech	Kemp (now Hello Robot CTO)	Core Stretch design, Healthcare Robotics Lab
University of Washington	Cakmak, Srinivasa	Tele-operation, accessible interfaces (CREATE lab)
UIUC	Rogers, Mahajan	Aging/disability support, participatory design (LIFE Home)
Meta FAIR	Chintala, Paxton	HomeRobot benchmark, RT-X, foundation models

14. GazeBot Project: Repository Recommendations & Usage Guide

This section maps the Hello Robot Stretch GitHub ecosystem to the specific needs of **GazeBot** -- a head-gaze and sip-and-puff shared autonomy system for assistive robot manipulation, being developed at the UIUC McKechnie Family LIFE Home. GazeBot uses Hello Robot Stretch 3 + Meta Quest 3 to enable individuals with severe motor disabilities (ALS, high-level SCI, locked-in syndrome) to perform activities of daily living through head movements and oral motor control.

14.1 Primary Repositories (Must-Use)

stretch_ros2 -- Core Robot Control Layer

GazeBot Role: Foundation of the entire robot control stack. Every GazeBot command ultimately flows through stretch_ros2.

Specific packages needed:

Package	GazeBot Usage
stretch_core	Main driver node in 'position' and 'navigation' modes. The stretch_driver publishes /mode, /battery, /is_homed topics your Unity app monitors. /runstop service is the E-stop backend for hard-puff.
stretch_nav2	Autonomous navigation backbone. User looks at destination via head-gaze -> Unity converts gaze ray + depth to 3D world coordinate -> sends Nav2 goal via Simple Commander API. SLAM mapping (offline_mapping.launch.py) creates the LIFE Home map. navigation.launch.py runs AMCL localization.
stretch_description	URDF for robot visualization in MR overlay. Could render robot state in Quest 3 via WebSocket.
stretch_funmap	Whole-body grasping planner. /funmap/trigger_reach_until_contact and /funmap/trigger_head_scan services are directly usable for GazeBot's manipulation pipeline.
stretch_demos	/handover_object/trigger_handover_object service is the "bring to me" action -- user sips to confirm, robot autonomously brings grasped object to wheelchair position.
stretch_simulation	MuJoCo sim for development when Stretch is unavailable. RoboCasa kitchen environments approximate LIFE Home layout.
hello_helpers	HelloNode class for rapid prototyping of new GazeBot ROS 2 nodes (head-gaze target converter, sip-puff action mapper).

stretch_ai -- AI & Perception Layer

GazeBot Role: Powers the shared autonomy intelligence -- object detection, grasp planning, and semantic understanding.

Critical components:

Component	GazeBot Usage
OWLv2 + SAMv2 perception	Object detection and segmentation when user's head-gaze ray highlights an object. User looks at cup -> gaze ray intersects with detected object bounding box -> visual confirmation in MR overlay.
AnyGrasp / grasp planning	Generates collision-free 6-DOF gripper poses after user sips to confirm object selection. Grasp preview shown in MR view before execution approval.
RobotClient API	High-level commands (move_to_nav_posture, move_to_manip_posture, arm_to) simplify GazeBot's control logic. ZMQ bridge enables offloading AI inference to GPU workstation.

Component	GazeBot Usage
stretch.app.grasp_object	Reference implementation for single-object grasping by name -- directly adaptable for GazeBot's 'Manipulate' mode.
stretch.app.mapping	Autonomous 3D mapping of the LIFE Home environment for navigation and semantic understanding.
DynaMem	Dynamic memory tracks objects that move (e.g., user asks robot to fetch cup that was moved). Essential for real home environments where objects are not static.
Whisper STT / TTS	Audio feedback via Bluetooth earbuds -- confirmation tones, status announcements, error alerts. Can also enable optional voice commands for users who retain speech.

stretch_body -- Low-Level Hardware Access

GazeBot Role: Direct hardware control beneath ROS 2, especially for custom behaviors.

Key uses:

- **Emergency stop implementation:** Hard puff -> robot.stop() halts all motion immediately. Monitor robot.pimu.status for runstop state.
- **Head pan/tilt control:** robot.head.move_to('head_pan', rad) maps directly to user's head-gaze yaw in 'Look' mode. Sub-degree precision matches Quest 3's 90 Hz head tracking.
- **Gripper control:** robot.end_of_arm.move_to('stretch_gripper', pct) for grasp execution after shared autonomy approval.
- **Battery monitoring:** robot.pimu.status battery data displayed in MR HUD (head-up display).
- **Sensor access:** Cliff sensors, IMU, wrist accelerometer for safety monitoring and contextual feedback to user.

14.2 Secondary Repositories (Highly Recommended)

stretch_tutorials -- Learning & Rapid Prototyping

Week 1 resource. Follow Track C ROS 2 tutorials in order: (1) Robot Driver, (2) Teleop, (3) Nav2 Stack, (4) Follow Joint Trajectory, (5) HelloNode, (6) Deep Perception. The 'Offloading Computation' tutorial is critical for GazeBot's GPU workstation architecture. Voice-to-Text and Voice Teleop examples inform audio feedback implementation.

stretch_web_teleop -- WebRTC Video Streaming

GazeBot Role: Provides the **WebRTC video streaming infrastructure** that GazeBot's Unity app builds upon. The proposal explicitly references this: 'WebRTC (via stretch_web_teleop infrastructure)'. Streams Stretch head camera and gripper camera feeds to Quest 3 via WebRTC. Also serves as the baseline comparison system for Week 7 evaluation (keyboard/mouse control vs. GazeBot).

stretch_dex_teleop -- Data Collection for Future Learning

Future direction. If GazeBot evolves to include adaptive autonomy (Section 11, Future Directions), dex_teleop enables collecting manipulation demonstrations for training imitation learning policies (ACT, Diffusion Policy via LeRobot). The \$295 teleop kit is already compatible with Stretch 3.

14.3 Reference Repositories

Repository	GazeBot Relevance
stretch_tool_share	The adapted_spoon_V1 tool is directly relevant to assistive feeding. The swivel_tablet_mount could hold a Quest 3 charging dock. Custom tools could be designed for specific ADL tasks (door opener, light switch pusher).
stretch_community	CMU WeHelp (shared autonomy for wheelchair users) and UW stretch-web-app (enhanced teleoperation) are directly related prior work. The Meta AI Habitat2 sim environment enables large-scale testing.
stretch_hardware_guides	Essential reference for understanding physical limits, wiring, and maintenance of the LIFE Home's Stretch robot.
stretch_pyfunmap	Python-native FUNMAP implementation -- alternative to ROS 2 funmap package for simpler integration with custom Python scripts.
stretch_simulation (MuJoCo)	Development and testing when robot is booked. RoboCasa kitchen environments approximate LIFE Home layout.

14.4 Recommended Integration Architecture

Data flow for a typical GazeBot 'fetch object' task:

Step	User Action	Software Component	Repository
1	User looks at object (head-gaze ray)	Unity/OpenXR app -> WebSocket	Custom (GazeBot)
2	Object detection & highlighting	OWLv2 + SAMv2 perception pipeline	stretch_ai
3	User sips to confirm object	Sip-and-puff -> Xbox Adaptive -> Quest 3	Custom (GazeBot)
4	Grasp planning & preview	AnyGrasp + FUNMAP	stretch_ai + stretch_ros2
5	User sips to approve grasp	MR overlay shows trajectory preview	Custom (GazeBot)
6	Robot executes grasp	joint_trajectory_server + stretch_driver	stretch_ros2 + stretch_body
7	User looks at destination (self)	Head-gaze -> 3D coordinate conversion	Custom (GazeBot)
8	Robot navigates to user	Nav2 Simple Commander API	stretch_ros2 (stretch_nav2)
9	Robot hands over object	/handover_object service	stretch_ros2 (stretch_demos)
10	Audio confirmation	Whisper TTS via Bluetooth earbuds	stretch_ai (audio module)

14.5 Repository Usage by Project Week

Week	Task	Primary Repos	Specific Files/Modules
1	Environment Setup	stretch_ros2, stretch_body, stretch_ai, stretch_tutorials	stretch_driver.launch.py, Track C tutorials, install.sh
2	Head-Gaze Control	stretch_body, stretch_web_teleop	robot.head.move_to(), WebRTC streaming, head pan/tilt API
3	Sip-and-Puff + Modes	stretch_body, stretch_ros2	robot.pimu (runstop for e-stop), /mode topic, HelloNode class
4	Autonomous Navigation	stretch_ros2 (stretch_nav2)	offline_mapping.launch.py, navigation.launch.py, Nav2 Simple Commander

Week	Task	Primary Repos	Specific Files/Modules
5	Shared Autonomy Manipulation	stretch_ai, stretch_ros2 (stretch_funmap)	grasp_object app, OWLv2+SAMv2, /funmap/trigger_* services
6	Task Presets + Audio	stretch_ai, stretch_ros2 (stretch_demos)	Whisper TTS, /handover_object, DynaMem for dynamic environments
7	Evaluation	stretch_web_teleop (baseline), all above	Web teleop as keyboard/mouse baseline comparison system
8	Documentation + Demo	stretch_community (reference)	Community portal for publishing GazeBot as open-source contribution

14.6 Repository-Specific Risks & Mitigations

Risk	Repository	Mitigation
stretch_ai requires GPU workstation with RTX 4090	stretch_ai	Use WiFi Access Point (\$995) for reliable connection. DynaMem has CPU-only mode (CLIP ViT-B/16).
AnyGrasp is closed-source	stretch_ai	Fallback to FUNMAP whole-body grasping (/funmap/trigger_reach_until_contact) which is fully open-source.
MuJoCo sim doesn't perfectly match real robot	stretch_ros2 simulation	Use MuJoCo for nav/logic testing only; validate manipulation on real robot during booked sessions.
ROS 2 bridge adds latency to AI pipeline	stretch_ai (ros2_bridge)	Profile ZMQ bridge latency early (Week 1). If >100ms, consider running AI models directly on robot NUC 12.
Nav2 may struggle in cluttered LIFE Home	stretch_ros2 (stretch_nav2)	Tune costmap inflation radius (default 0.20m). Use RTAB-Map for 3D visual navigation as fallback.

14.7 Most Relevant Publications for GazeBot

Paper	GazeBot Relevance
Immersive Participatory Design (Olatunji, Mahajan et al., 2024)	Direct predecessor at LIFE Home. Same PI (Mahajan), same robot, same mission. GazeBot extends to users who cannot use hands or voice.
HAT: Head-Worn Assistive Teleop (CMU, 2023)	Most directly comparable system: head-worn interface for Henry Evans. GazeBot improves on this with XR integration and shared autonomy.
Bimanual HD-EMG Control (CMU, 2026)	Alternative input modality (muscle signals vs. head-gaze). 12-day in-home study methodology is a model for GazeBot's future longitudinal deployment.
Accessible Remote Tele-operation (UW, 2021)	Browser-based accessible interface study. Informs GazeBot's usability evaluation design (NASA-TLX, SUS).
AccessTeleopKit (UW, UIST 2024)	Open-source accessible tele-operation toolkit. Could complement GazeBot as fallback interface.
OK-Robot (NYU/Meta, 2024)	Zero-shot pick-and-drop architecture. GazeBot could adopt this modular approach for manipulation.
ForceSight (Hello Robot, ICRA 2024)	Text-guided manipulation with visual-force goals. Could enhance GazeBot's grasping with force feedback.
Shared Autonomy with Learned Latent Actions (Jeon, RSS 2020)	Theoretical foundation for GazeBot's shared autonomy paradigm. 2-DOF input controlling 7-DOF robot.

15. Sources & References

Hello Robot GitHub Organization

<https://github.com/hello-robot>

stretch_ai Repository

https://github.com/hello-robot/stretch_ai

stretch_ros2 Repository

https://github.com/hello-robot/stretch_ros2

stretch_ros Repository

https://github.com/hello-robot/stretch_ros

stretch_body Repository

https://github.com/hello-robot/stretch_body

stretch_firmware Repository

https://github.com/hello-robot/stretch_firmware

stretch_tutorials Repository

https://github.com/hello-robot/stretch_tutorials

stretch_dex_teleop Repository

https://github.com/hello-robot/stretch_dex_teleop

stretch_tool_share Repository

https://github.com/hello-robot/stretch_tool_share

stretch_community Repository

https://github.com/hello-robot/stretch_community

Stretch Documentation (v0.3)

<https://docs.hello-robot.com/0.3/>

Stretch Documentation (v0.2)

<https://docs.hello-robot.com/0.2/>

Stretch AI Product Page

<https://hello-robot.com/stretch-ai>

Stretch 3 Product Page

<https://hello-robot.com/stretch-3-product>

Stretch Community Resources

https://docs.hello-robot.com/0.3/getting_started/community_resources/

Stretch Forum

<https://forum.hello-robot.com>

The Design of Stretch (arXiv:2109.10892)

<https://arxiv.org/abs/2109.10892>

DynaMem Paper (arXiv:2411.04999)

<https://arxiv.org/abs/2411.04999>

OK-Robot (arXiv:2401.12202)

<https://arxiv.org/abs/2401.12202>

ForceSight (arXiv:2309.12312)

<https://arxiv.org/abs/2309.12312>

HomeRobot OVMM (arXiv:2306.11565)

<https://arxiv.org/abs/2306.11565>

HAT Interface (arXiv:2312.15071)

<https://arxiv.org/abs/2312.15071>

Robot Utility Models (arXiv:2409.05865)

<https://arxiv.org/abs/2409.05865>

Harmonic Mobile Manipulation (arXiv:2312.06639)

<https://arxiv.org/abs/2312.06639>

PoliFormer (arXiv:2406.20083)

<https://arxiv.org/abs/2406.20083>

SPOC (CVPR 2024)

<https://spoc-robot.github.io/>

Dobb-E (arXiv:2311.16098)

<https://arxiv.org/abs/2311.16098>

Open X-Embodiment (arXiv:2310.08864)

<https://arxiv.org/abs/2310.08864>

Bimanual EMG Control (arXiv:2602.02773)

<https://arxiv.org/abs/2602.02773>

IEEE Spectrum: Stretch 3 Review

<https://spectrum.ieee.org/hello-robot-stretch-3>

Introducing Stretch AI (Chris Paxton Substack)

<https://itcanthink.substack.com/p/introducing-stretch-ai>